BEN KROTONSKY

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EDUCATION

Northern Arizona University — **Flagstaff, Arizona**
Bachelor of Science, Mechanical Engineering, Minor in German | GPA: 4.0 / 4.0 — *Expected May 2027*

WORK EXPERIENCE

Dynamic and Active Systems Lab (DASL) — NAU — **September 2023 – Present**
Undergraduate Research Assistant — *Flagstaff, Arizona*

- Built orthomosaic stitching workflow in Agisoft Metashape and WebODM for aerial thermal analysis of underground water pipelines.
- Executed 20+ drone test missions using a DJI Mavic 3T across eight sites to acquire thermal imagery of known leaks and establish POI detection criteria.
- Developed a statistical temperature variance model to distinguish anomalous thermal signatures from background population noise.
- Created an automated POI identification algorithm achieving 66% detection accuracy without human interference.
- Conducted moisture analyses at two field sites to characterize surface leak behavior and refine detection thresholds.
- Designed a GUI to streamline the thermal analysis interface, reducing total processing time by 37% versus the original workflow.

match — Leibniz University Hannover (LUH) — **November 2024 – Present**
DAAD RISE Research Intern — *Hannover, Germany*

- Developed a geometry-based algorithm to quantify stable pose probabilities for arbitrary workpieces using Centroid Solid Angle (CSA) analysis.
- Extracted part geometry from 3D models, identified valid resting positions, filtered unstable poses, and constructed centroid-to-face prisms to compute solid angle proportions.
- Validated algorithm across 5 parts with 5–6 resting positions each, achieving a mean absolute error of 3.92% against simulation — without requiring large sample sizes.
- Results will be integrated into an aerodynamic manufacturing chute to optimize workpiece orientation sequences.
- Approach eliminates reliance on extensive physical testing or large-sample simulation required by existing probabilistic methods.

ENGINEERING PROJECTS

Steady-State Pin Fin Heat Transfer Algorithm — **NAU — Heat Transfer**

- Developed a generalized algorithm modeling conductive-convective heat transfer across arbitrary cross-sectional geometries including conical and cylindrical fins.
- Implemented configurable boundary conditions: insulated tips, convective ends, and prescribed base temperatures.
- Applied adaptive nodal analysis, iterating resolution until convergence criteria were satisfied to produce accurate temperature distributions and heat transfer rate predictions.

Formula SAE Front Wing — Active Aero System — **NAU — Engineering Design**

- Served as Data Analysis Lead; selected airfoil profile and determined optimal angles of attack for cornering (maximum downforce) and straights (minimum drag).
- Conducted aerodynamic force analysis across a 0°–20° angle-of-attack sweep to identify optimal configurations.
- Designed wing produces approximately 76 N of downforce in cornering with near-zero drag on straights.
- Pioneered NAU Formula SAE's first aerodynamic package, delivering a significant performance gain over the team's previously unaired configuration.

TECHNICAL SKILLS

Engineering Software: MATLAB, SolidWorks, Autodesk Inventor, Agisoft Metashape, WebODM
Data & Analysis: Microsoft Excel, MATLAB (numerical methods, nodal analysis, statistical modeling)
Field & Hardware: Drone Operation (DJI Mavic 3T, RTK), Thermal IR Imaging
Methods: Finite Difference Analysis, Aerodynamic Analysis, Probabilistic Modeling, Algorithm Development, GUI Design
Languages: English (Native), German (Professional Working Proficiency, B2)